

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA0304

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A **Application - Based Questions**

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**
2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**
3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. **(5 Marks)**
4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**
5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. **(5 Marks)**

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. **(5 Marks)**

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. **(5 Marks)**

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

Code of Conduct:

*I **K Varshan** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. Reinforcement Learning in a Ride-Sharing Company

RL Framework

State Space (S):

- Driver location
- Passenger location
- Traffic conditions
- Driver availability
- Time of day

Action Space (A):

- Assign a driver to a passenger
- Reject request
- Wait for a better driver
- Reassign another driver

Reward Function (R):

- **+10** for quick pickup
- **+20** for successful ride completion
- **-10** for long passenger waiting time
- **-15** for ride cancellation

Working:

The RL agent learns which driver should be assigned to each passenger. By receiving rewards for fast pickups and successful trips, it gradually minimizes waiting time and improves customer satisfaction.

2. RL in an E-Learning Platform

Exploration and Exploitation

Exploration:

- Recommends new study materials.
- Helps discover better learning resources.
- Prevents repetitive recommendations.

Exploitation:

- Recommends materials the student already prefers.
- Improves learning efficiency.
- Uses past learning history.

Importance:

Balancing exploration and exploitation ensures students receive both familiar and new educational content, improving engagement and learning performance.

3. RL for a Robotic Vacuum Cleaner

RL Framework

State Space

- Robot position
- Dirt locations
- Battery level
- Obstacle positions

Action Space

- Move forward
- Turn left
- Turn right

- Clean
- Return to charging station

Policy

The policy decides the best action in each state to maximize cleaning efficiency while avoiding obstacles.

Value Function

The value function estimates the long-term reward of each state. As learning continues, the robot identifies efficient cleaning paths, reducing time and energy consumption.

4. RL in Cryptocurrency Trading

Actions

- Buy
- Sell
- Hold

Reward Design Challenges

- Delayed profits
- Highly volatile market
- Transaction costs
- Risk management

Poor Reward Design Effects

- Excessive buying or selling
- Large financial losses
- Overfitting to market history
- Unstable trading decisions

A well-designed reward function encourages long-term profit while minimizing risk.

5. RL in a Smart Grid System

RL Framework

State Space

- Electricity demand
- Power generation
- Battery storage level
- Grid status

Action Space

- Increase power supply
- Reduce supply
- Store excess electricity
- Distribute electricity

Reward Function

- **+20** for balanced power distribution
- **+15** for reducing energy waste
- **-20** for power outages
- **-10** for excessive energy consumption

The RL agent learns efficient electricity management to reduce costs and maintain grid stability.

6. RL in a Healthcare Monitoring System

RL Framework

State Space

- Patient health condition
- Blood pressure
- Heart rate
- Previous medication dosage

Action Space

- Increase dosage
- Decrease dosage
- Maintain dosage
- Recommend medical review

Reward Function

- **+20** for improved patient health
- **+10** for stable condition
- **-20** for adverse side effects
- **-30** for health deterioration

Learning Improvement

The RL agent learns the most effective dosage adjustments over time, improving treatment outcomes and reducing complications.

7. RL in a Drone Delivery System

Exploration

- Tests new delivery routes
- Learns alternative paths
- Adapts to changing weather and traffic

Exploitation

- Uses previously successful routes
- Reduces delivery time
- Saves battery power

Effect on Route Optimization

Balancing exploration and exploitation enables the drone to continuously discover efficient routes while maintaining reliable deliveries.

8. RL in a Manufacturing Robot

RL Framework

State Space

- Robot position
- Assembly stage
- Tool status
- Product condition

Action Space

- Pick component
- Place component
- Tighten screw
- Inspect product

Policy

The policy selects the best assembly action at each stage to maximize production efficiency.

Value Function

The value function estimates future rewards, helping the robot choose actions that reduce assembly errors and production time.

9. RL in an Online Advertising System

Reward Design Challenges

- Delayed customer purchases
- Click fraud
- Multiple user preferences
- Balancing clicks and revenue

Poor Reward Design Effects

- Too many irrelevant advertisements
- Reduced click-through rate (CTR)
- Lower customer satisfaction
- Reduced advertising revenue

A proper reward function encourages meaningful user engagement rather than only maximizing clicks.

10. RL in a Smart Irrigation System

RL Framework

State Space

- Soil moisture
- Weather conditions
- Temperature
- Water availability
- Crop growth stage

Action Space

- Start irrigation
- Stop irrigation
- Increase water flow
- Reduce water flow

Reward Function

- **+20** for healthy crop growth
- **+15** for water conservation
- **-20** for overwatering
- **-20** for underwatering

The RL agent learns the optimal irrigation schedule that maximizes crop yield while minimizing water usage.